Air, Reimagined

Smart ventilation for exhaust fans

Sense locally. Show clearly.

team AiroRhythm

01 | PROBLEM

Ventilation should respond to the room.

THE GAP

Moisture, smoke-like events and motor drag can go unseen, leaving ventilation and maintenance to manual attention.

THE PROPOSAL

AiroRhythm adds ESP32 sensing and control to a compatible Atomberg exhaust fan, linking humidity automation, smoke alerts and motor-load monitoring to one app.

FROM SIGNAL TO SERVICE

The real fan stays central

1 SENSE
Humidity / smoke / motor load

2 DECIDE
ESP32 filters signals; applies local rules

3 VENTILATE
Safe commands reach the actual fan

4 INFORM
App status, alerts and controls

LOCAL-FIRST

Humidity rules can run without cloud access.

HUMAN-IN-CONTROL

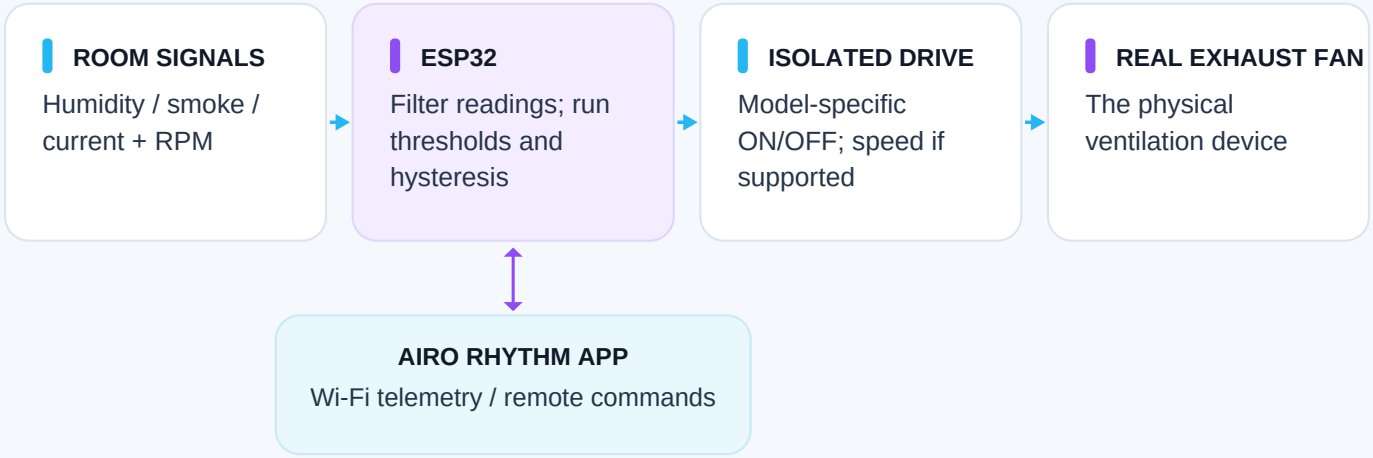
Keep the physical switch and fan protections.

PROTOTYPE SCOPE

The Flutter prototype simulates readings and fan state. Its virtual fan is status-only; the real exhaust fan ventilates. Hardware integration is future work.

How the system works

Local intelligence connects sensors, app and fan



TWO APP MODES

DISPLAY | MONITOR

Live humidity, smoke, load, fan status, last update and events; graphic mirrors reported state.

REMOTE | CONTROL + AUTOMATE

Manual ON/OFF, speed if supported, thresholds and schedules; confirm commands and override automation.

SENSOR-TO-ACTION GUIDE

SIGNAL	WHAT IT MEANS	SYSTEM RESPONSE
Humidity sensor	Moisture crosses a set point.	Start fan; a lower OFF threshold and minimum runtime prevent cycling.
Smoke / air-quality sensor	Smoke-like reading.	Alert locally and in app; configurable response, not a certified alarm.
Rotation-load proxy	Motor current/RPM versus clean baseline.	Current rise or RPM loss prompts inspection; indicator, not diagnosis.

SAFETY AND RESILIENCE

Use isolated, model-specific control; retain the wall switch and protections. Stale readings disable dependent rules and alert. Humidity runs locally; smoke sensing is not life-safety protection.

From concept to a measurable pilot

AiroRhythm

Build in stages; prove safety and value before scaling.

FEASIBILITY

ESP32, humidity/smoke sensors and the Flutter app form a practical stack. Compare current and RPM to flag drag. Validate placement, calibration, isolated actuation and electrical safety.

VIABILITY

A retrofit or integrated model could automate without changing daily use. Local rules limit cloud dependence; the app supports setup and service prompts. Validate cost, installation, reliability, false alerts, energy use and demand before productization.

POTENTIAL BENEFITS | TO BE MEASURED

COMFORT

Humidity-led ventilation.

AWARENESS

Smoke alerts and live fan status.

CARE

Early prompt for inspection or cleaning.

EFFICIENCY

Hysteresis limits cycling; verify energy impact.

PROTOTYPE PATH

NOW | SOFTWARE

Flutter simulates readings and fan state; no hardware yet.

NEXT | BENCH

Add ESP32 and calibrated sensors; test stale data, override and isolated control.

PILOT | VALIDATE

Compare alerts, humidity response, drag flags and runtime against references.

PILOT SCORECARD

Energy savings remain unclaimed until measured.

Trigger accuracy / false alerts / response / drag agreement / command acknowledgement / runtime

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